Name: Dakshata Chnadrakant Salgaonkar

Branch: CSEDS

Division:SE3

Roll no:30

Experiment no:9

Aim: implement a search functionality to filter API results dynamically

Code:

```
import React, { useState, useEffect } from 'react';


function App() {
  // --- STATE ---
  const [users, setUsers] = useState([]); // Stores the original API data
  const [searchTerm, setSearchTerm] = useState(""); // Stores what the user
types
  const [loading, setLoading] = useState(true); // Shows a loader while fetching


  // --- SIDE EFFECT: FETCH DATA ---
  useEffect(() => {
    fetch('https://jsonplaceholder.typicode.com/users')
      .then(response => response.json())
      .then(data => {
        setUsers(data);
        setLoading(false);
```

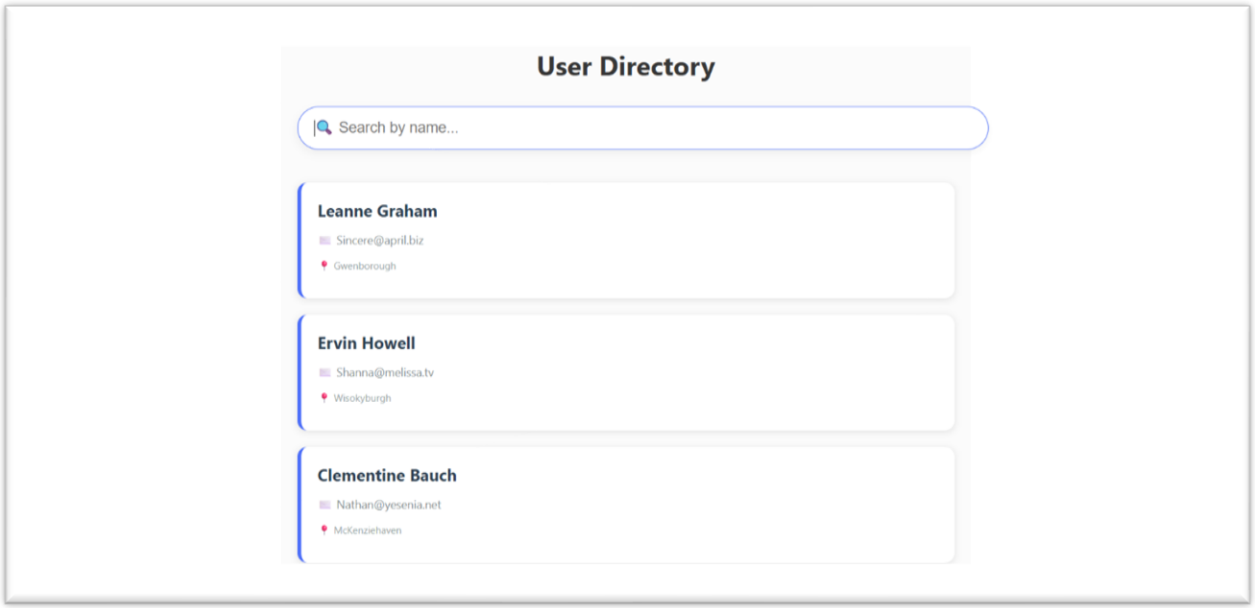
```
    })
    .catch(err => console.error("Error fetching data:", err));
}, []);

// --- LOGIC: DYNAMIC FILTERING ---
// This creates a sub-list of users whose names match the search term
const filteredUsers = users.filter(user =>
  user.name.toLowerCase().includes(searchTerm.toLowerCase())
);

// --- STYLING ---
const styles = {
  container: {
    padding: '40px 20px',
    maxWidth: '800px',
    margin: '0 auto',
    fontFamily: "'Segoe UI', Tahoma, Geneva, Verdana, sans-serif",
    backgroundColor: '#f9f9f9',
    minHeight: '100vh',
  },
  header: { textAlign: 'center', color: '#333', marginBottom: '30px' },
  searchBar: {
    width: '100%',
    padding: '15px 20px',
    fontSize: '18px',
```

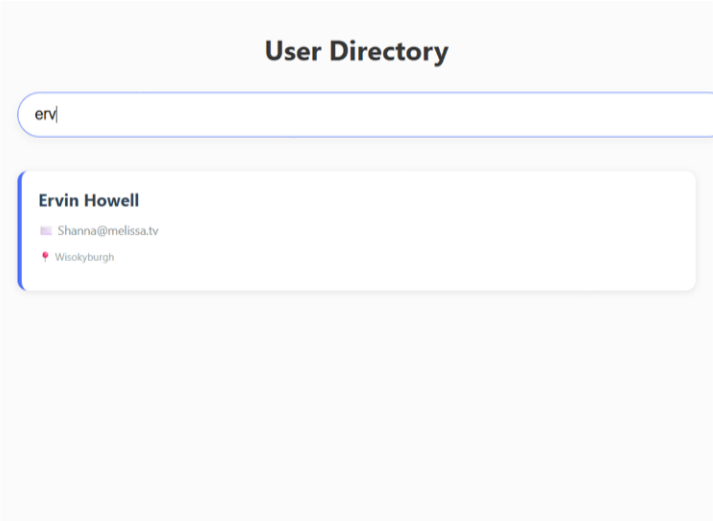
```
    borderRadius: '30px',
    border: '1px solid #ddd',
    boxShadow: '0 4px 10px rgba(0,0,0,0.05)',
    marginBottom: '40px',
    outline: 'none',
    transition: 'border 0.3s',
  },
  grid: {
    display: 'grid',
    gridTemplateColumns: 'repeat(auto-fill, minmax(250px, 1xl))',
    gap: '20px',
  },
  card: {
    backgroundColor: '#fff',
    padding: '20px',
    borderRadius: '12px',
    boxShadow: '0 2px 8px rgba(0,0,0,0.1)',
    borderLeft: '5px solid #4a6cf7',
  },
  name: { margin: '0 0 5px 0', color: '#2c3e50', fontSize: '20px' },
  email: { color: '#7f8c8d', fontSize: '14px' }
};

return (
  <div style={styles.container}>
```

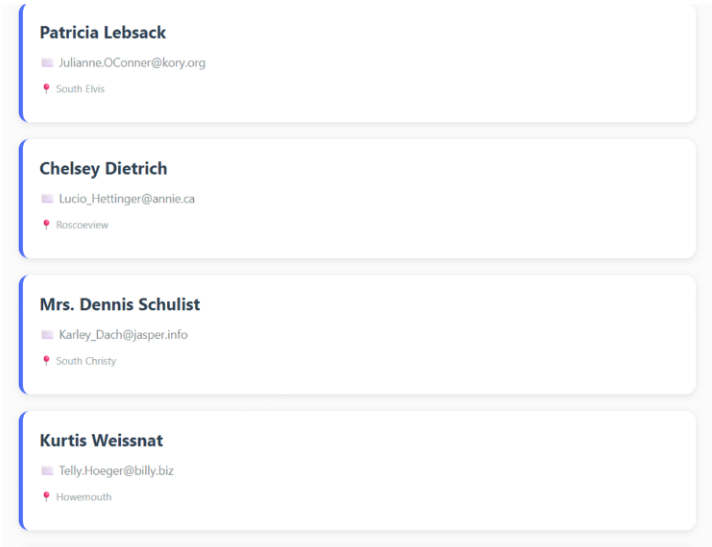
```
    <h1 style={styles.header}>User Directory</h1>

    {/* Search Input */}
    <input
      style={styles.searchBar}
      type="text"
      placeholder="🔍 Search by name..."
      value={searchTerm}
      onChange={(e) => setSearchTerm(e.target.value)}
      onFocus={(e) => e.target.style.borderColor = '#4a6cf7'}
    />

    {loading ? (
      <p style={{textAlign: 'center'}}>Loading Users...</p>
    ) : (
      <div style={styles.grid}>
        {filteredUsers.length > 0 ? (
          filteredUsers.map(user => (
            <div key={user.id} style={styles.card}>
              <h3 style={styles.name}>{user.name}</h3>
              <p style={styles.email}>📧 {user.email}</p>
              <p style={{fontSize: '12px', color: '#95a5a6'}}>📍
{user.address.city}</p>
            </div>
          ))
```

```
        ) : (
          <p style={{textAlign: 'center', gridColumn: '1/-1', color: '#e74c3c'}}>
            No users found matching "{searchTerm}"
          </p>
        )}
      </div>
    )}
  </div>
 );
}

export default App;
```

Output:




User Directory
erv
Ervin Howell
Shanna@melissa.tv
Wisokyburgh


Patricia Lebsack
Julianne.OConner@kory.org
South Elvis
Chelsey Dietrich
Lucio_Hettinger@annie.ca
Roscoeview
Mrs. Dennis Schulist
Karley_Dach@jasper.info
South Christy
Kurtis Weissnat
Telly.Hoeger@billy.biz
Howemouth